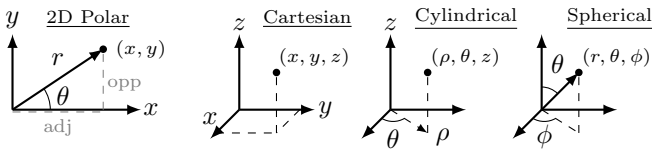


Coordinate Systems



Kinematics - Constant Acceleration

$$\vec{v}_f = \vec{v}_i + \vec{a}t \quad \vec{x}_f = \vec{x}_i + \vec{v}_i t + \frac{1}{2}\vec{a}t^2$$

$$v_f^2 = v_i^2 + 2a(x_f - x_i) \quad \vec{a} : \text{acceleration [m/s}^2\text{]}$$

$$\vec{a} = \frac{d\vec{v}}{dt} \quad \vec{v} = \frac{d\vec{x}}{dt} \quad \vec{v} : \text{velocity [m/s]}$$

$$v = \frac{\Delta x}{\Delta t} \quad (\text{only if } a = 0) \quad \vec{x} : \text{position [m]} \quad t : \text{time [s]}$$

Force F [Newton] [N] [kg · m/s²]

$$\sum \vec{F} = m\vec{a} \quad \vec{F}_{12} = -\vec{F}_{21} \quad (\text{by, on})$$

$$a_c = v^2/r \quad \text{Weight } mg \text{ down, center of Earth}$$

$$a = \sqrt{a_t^2 + a_c^2} \quad \text{Tension } T \text{ only pulls}$$

$$f = \mu N \quad \text{Normal } N \text{ perpendicular, reacts}$$

$$F_s = -kx \quad \text{Friction } f \text{ opposes motion}$$

$$\quad \quad \quad \text{Spring } F_s \text{ spring const } k \text{ [N/m]}$$

Energy E [Joule] [J]

$$W_{nc} = \Delta K + \Delta U \quad F_x = -\frac{dU}{dx} \quad W = \int \vec{F}(r) \cdot d\vec{r}$$

$$K = \frac{1}{2}mv^2 \quad \leftarrow \text{Kinetic E} \quad \mathbb{P} = \frac{dE}{dt} = \vec{F} \cdot \vec{v} = \vec{\tau} \cdot \vec{\omega}$$

$$K_{\text{Rot}} = \frac{1}{2}I\omega^2 \quad \text{Power } \rightarrow \quad [\text{Watt} = \text{J/s}]$$

$$U_g = mgh \quad \leftarrow \text{Potential E} \quad \vec{A} \cdot \vec{B} = AB \cos(\theta)$$

$$U_s = \frac{1}{2}kx^2 \quad \text{Dot Product } \rightarrow \quad = A_x B_x + A_y B_y + A_z B_z$$

Momentum p [kg · m/s]

$$\vec{p} = m\vec{v} \quad \vec{F} = \frac{d\vec{p}}{dt} \quad \Delta \vec{p}_{\text{sys}} = \int \vec{F}_{\text{ext}}(t) dt$$

$$m_1 \vec{v}_{1i} + m_2 \vec{v}_{2i} = m_1 \vec{v}_{1f} + m_2 \vec{v}_{2f} \quad (\leftarrow \Delta p = 0)$$

$$\frac{1}{2}m_1 |\vec{v}_{1i}|^2 + \frac{1}{2}m_2 |\vec{v}_{2i}|^2 = \frac{1}{2}m_1 |\vec{v}_{1f}|^2 + \frac{1}{2}m_2 |\vec{v}_{2f}|^2 \quad (\leftarrow \Delta K = 0)$$

$$v_{1i} + v_{1f} = v_{2i} + v_{2f} \quad (\leftarrow \text{1D elastic})$$

Angular Variables

thumb=arm × fingers ⊗ in
⊙ out

$$x = r\theta \quad \text{arm} \quad \text{thumb} \quad \vec{\omega}$$

$$v = r\omega \quad \text{fingers} \quad \text{motion}$$

$$a = r\alpha$$

$$\vec{\omega}_f = \vec{\omega}_i + \vec{\alpha}t \quad \vec{\theta} = \vec{\theta}_i + \vec{\omega}_i t + \frac{1}{2}\vec{\alpha}t^2$$

θ	ang pos [rad]	x	A	amplitude	
ω	ang vel [rad/s]	v	f	frequency [Hz]	$\omega = 2\pi f$
α	ang acc [rad/s ²]	a	λ	wavelength [m]	
τ	torque ↔ force	F	k_λ	wave number	$k_\lambda = 2\pi/\lambda$
I	MoI ↔ mass	m	ϕ	phase angle	
L	ang momentum	p	T	period [s]	$T = 1/f$

Torque τ [N · m] Angular Momentum L [kg · m²/s]

$$\sum \vec{\tau} = I\vec{\alpha} \quad \vec{\tau} = \vec{r} \times \vec{F} \quad |\vec{\tau}| = rF \sin \theta$$

$$\vec{L} = I\vec{\omega} \quad \vec{L} = \vec{r} \times \vec{p} \quad |\vec{A} \times \vec{B}| = AB \sin \theta$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \leftarrow \text{Cross Product}$$

down right(+)
down left(-)

Center of Mass [m]

$$x_{\text{CM}} = \frac{\sum x_i \cdot m_i}{\sum m_i} \quad \vec{r}_{\text{CM}} = \frac{1}{M} \int \vec{r} dm = \frac{1}{M} \iiint \rho(\vec{r}) \vec{r} dV$$

Moment of Inertia [kg · m²]

$$I = \int r^2 dm \quad I_{\parallel} = I_{\text{CM}} + MD^2$$

$$\rho = m/V \quad \sigma = m/A \quad \lambda = m/\ell$$

Shape	MoI
Point Mass	MR^2
Thin Rod (center)	$1/12 \cdot ML^2$
Thin Rod (end)	$1/3 \cdot ML^2$
Rectangular Plate	$1/12 \cdot M(L_1^2 + L_2^2)$
Cylindrical Shell	MR^2
Solid Cylinder	$1/2 \cdot MR^2$
Hollow Cylinder	$1/2 \cdot M(R_1^2 + R_2^2)$
Spherical Shell	$2/3 \cdot MR^2$
Solid Sphere	$2/5 \cdot MR^2$
Hollow Sphere	$2/5 \cdot \frac{R_1^5 - R_2^5}{R_1^3 - R_2^3}$

Universal Gravitation

$$F_G = G \frac{m_1 m_2}{r^2} \quad U_G = -G \frac{m_1 m_2}{r} \quad v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$$

Fluid Dynamics - Pressure P [Pascal] [Pa] [N/m²]

$$F = \int P dA \quad P = P_0 + \rho gh \quad F_B = \rho g V_{\text{displ}}$$

$$A_1 v_1 = A_2 v_2 = \text{const} \quad P + \frac{1}{2} \rho v^2 + \rho gy = \text{const}$$

Oscillations - 2nd order ODE

$$\frac{d^2 x}{dt^2} = -\omega^2 x \Rightarrow x(t) = A \cos(\omega t + \phi)$$

$$m \frac{d^2 x}{dt^2} = -b \frac{dx}{dt} - kx \Rightarrow x(t) = A e^{-\omega_d t} \cos(\omega t + \phi)$$

Simple	AngFreq	Damped	Behavior
mass-spring	$\omega = \sqrt{k/m}$	$\omega_0 = \sqrt{k/m}$	$\omega_d < \omega_0$ under
pendulum	$\omega = \sqrt{g/\ell}$	$\omega_d = b/2m$	$\omega_d = \omega_0$ critical
rigid pend	$\omega = \sqrt{\frac{mgd}{I}}$	$\omega^2 = \omega_0^2 - \omega_d^2$	$\omega_d > \omega_0$ over

Travelling Waves - 2nd order PDE

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2} \Rightarrow y(x, t) = f(x \mp vt)$$

$$y(x, t) = A \sin(k_\lambda x - \omega t + \phi) \quad v = \lambda f = \sqrt{\frac{F_T}{m/\ell}}$$

$$v_t = \partial y / \partial t = -A\omega \cos(\sim) \quad a_t = \partial^2 y / \partial t^2 = -A\omega^2 \sin(\sim)$$

Mechanical Waves

$$f' = \left(\frac{v + v_o}{v - v_s} \right) f \quad f_{\text{beat}} = |f_1 - f_2|$$

$$f_n = \frac{(2n)v}{4L} \quad \leftarrow \begin{matrix} \text{fixed/fixed} \\ \text{fixed/free} \rightarrow \end{matrix} \quad f_n = \frac{(2n+1)v}{4L}$$

Physics - Mechanics

Conversions

$1\text{ in} = 2.54\text{ cm}$	$1\text{ ft}\cdot\text{lb} = 1.356\text{ N}\cdot\text{m}$	tera	T	10^{12}
$1\text{ ft} = 0.3048\text{ m}$	$1\text{ cal} = 4.186\text{ J}$	giga	G	10^9
$1\text{ yard} = 3\text{ ft}$	$1\text{ kWh} = 3.60\text{ MJ}$	mega	M	10^6
$1\text{ mi} = 5280\text{ ft}$	$1\text{ hp} = 746\text{ J/s}$	kilo	k	10^3
$1\frac{\text{mi}}{\text{hr}} = 0.447\frac{\text{m}}{\text{s}}$	$1\text{ C} = 6.25 \cdot 10^{18}\text{ e}$	centi	c	10^{-2}
$1\text{ L} = 1000\text{ cm}^3$	$1\text{ eV} = 1.60 \cdot 10^{-19}\text{ J}$	milli	m	10^{-3}
$1\text{ gal} = 3.786\text{ L}$	$T_C = 5/9(T_F - 32)$	micro	μ	10^{-6}
$1\text{ ft}^3 = 28.32\text{ L}$	$T = T_C + 273$	nano	n	10^{-9}
$1\text{ lb} = 4.448\text{ N}$	$1\text{ atm} = 101325\text{ Pa}$	pico	p	10^{-12}

Greek Symbols

alpha	α	iota	ι	rho	ρ
beta	β	kappa	κ	sigma	σ Σ
gamma	γ Γ	lambda	λ Λ	tau	τ
delta	δ Δ	mu	μ	upsilon	ν Υ
epsilon	ϵ	nu	ν	phi	ϕ Φ
zeta	ζ	xi	ξ Ξ	chi	χ
eta	η	omicron	$\omicron$	psi	ψ Ψ
theta	θ Θ	pi	π Π	omega	ω Ω

Physical Constants

Name	Symbol	Value
Gravitation Constant	G	$6.67 \cdot 10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2$
Earth Surface Accel	g	9.8 m/s^2
Boltzmann Constant	k_B	$1.38 \cdot 10^{-23}\text{ J/K}$
Avogadros Number	N_A	$6.02 \cdot 10^{23}\text{ mol}^{-1}$
Ideal Gas Constant	R	$8.314\text{ J}/(\text{mol}\cdot\text{K})$
Stefan-Boltzmann	σ	$5.67 \cdot 10^{-8}\text{ W}/(\text{m}^2\cdot\text{K}^4)$
Speed of Light	c	$2.99792458 \cdot 10^8\text{ m/s}$
Electrostatic Constant	k_e	$8.99 \cdot 10^9\text{ N}\cdot\text{m}^2/\text{C}^2$
Electric Permittivity	ϵ_0	$8.854 \cdot 10^{-12}\text{ C}^2/(\text{N}\cdot\text{m}^2)$
Magnetic Permeability	μ_0	$4\pi \cdot 10^{-7}\text{ T}\cdot\text{m/A}$
Elementary Charge	e	$1.60 \cdot 10^{-19}\text{ C}$
Planck Constant	h	$6.63 \cdot 10^{-34}\text{ J}\cdot\text{s}$
Rydberg Constant	R_H	$1.09737 \cdot 10^{-7}\text{ m}^{-1}$
Bohr Magneton	μ_B	$9.274 \cdot 10^{-24}\text{ J/T}$
Bohr Radius	a_0	$5.292 \cdot 10^{-11}\text{ m}$
Rest mass of Electron	m_e	$9.1093 \cdot 10^{-31}\text{ kg}$
Rest mass of Proton	m_p	$1.6726 \cdot 10^{-27}\text{ kg}$
Rest mass of Neutron	m_n	$1.6749 \cdot 10^{-27}\text{ kg}$
Atomic Mass Unit	u	$931.49\text{ MeV}/c^2$
Mass Earth	M_E	$5.97 \cdot 10^{24}\text{ kg}$
Mass Sun	$M_\odot$	$1.99 \cdot 10^{30}\text{ kg}$
Radius Earth	R_E	$6.37 \cdot 10^6\text{ m}$
Radius Sun	$R_\odot$	$6.96 \cdot 10^8\text{ m}$
Distance Earth/Sun	AU	$1.496 \cdot 10^{11}\text{ m}$

Speed of Sound [m/s]

Material	v	Material	v	Material	v
Air 25°C	343	Ice	3845	Copper	5010
Air 0°C	331	Glass	5640	Aluminum	6420
Steam	478	P-wave	4k-7k	Iron	5950
Water	1492	S-wave	2k-5k	Lead	1960

Friction Coefficients $f = \mu N$

Material	μ_k	μ_s	Material	μ_k	μ_s
Tire-DryConc	0.68	0.90	Glass-Glass	0.4	0.94
Tire-WetConc	0.58		Wood-Wood	0.30	0.42
Tire-Ice	0.15		Ski-Snow	0.05	0.14
Steel-Steel	0.57	0.74	Teflon-Teflon	0.04	0.04
Copper-Steel	0.36	0.53	Synovial Fluid	0.01	0.01

Equation of State - P [Pa] V [m³] T [K]

$$PV = nRT \quad PV = Nk_B T \quad \text{Ideal Gas}$$

$$P = \frac{nRT}{V-b} - \frac{a}{V^2} \quad \text{Van der Waals} \quad \begin{matrix} a \rightarrow \text{strength} \\ b \rightarrow \text{volume} \end{matrix}$$

Statistical Thermodynamics Ideal Gas

$$\Delta U = n C_V dT \quad P = \frac{2}{3} \frac{N}{V} \left(\frac{1}{2} m \bar{v}^2 \right)$$

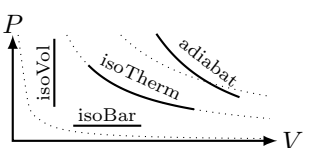
$$C_V = \frac{\text{DoF}}{2} R \quad C_P = \frac{\text{DoF} + 2}{2} R \quad \gamma = C_P / C_V$$

$$S = k_B \ln(\mathbb{P}) \quad \mathbb{P}_{\text{Boltz}} \propto \exp(-E_i / k_B T)$$

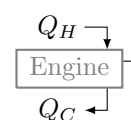
1st Law of Thermodynamics [J] [Joule]

$$\Delta U = \Delta Q + \Delta W \quad dU = TdS + (-PdV)$$

$$\Delta W = \int_{V_i}^{V_f} -P dV \Rightarrow$$

$$\Delta Q = \text{“Heat”} = \int TdS$$


Heat Engine



$$W_{\text{eng}} = |Q_H| - |Q_C|$$

$$e = \frac{W_{\text{eng}}}{Q_H} = 1 - \frac{|Q_C|}{|Q_H|}$$

$$e_{\text{carnot}} = 1 - \frac{T_C}{T_H}$$

PV^m	Process	ΔU_{int}	ΔQ	ΔW
	General	$nC_V dT$	$dU - dW$	$-\int P dV$
∞	Isovolume	$nC_V \Delta T$	$nC_V \Delta T$	0
0	Isobaric	$nC_V \Delta T$	$nC_P \Delta T$	$-P \Delta V$
1	Isothermal	0	$-\Delta W$	$-nRT \ln\left(\frac{V_f}{V_i}\right)$
γ	Adiabatic	$nC_V \Delta T$	0	$-\frac{(P_f V_f - P_i V_i)}{(1-\gamma)}$

$$P_1 V_1^\gamma = P_2 V_2^\gamma = \text{constant} \quad (\leftarrow \text{Adiabatic } dQ = 0)$$

Thermal Material Properties

Material	Density ρ (kg/m^3)	Specific Heat c ($\frac{\text{J}}{\text{kg}\cdot\text{C}^\circ}$)	Thermal Cond k ($\frac{\text{W}}{\text{m}\cdot\text{C}^\circ}$)	Volume Expans β ($\frac{1}{\text{C}^\circ}$) $\cdot 10^{-6}$
Steam	0.590	2010	0.027	3400
Air	1.29	1004	0.023	3670
Water	1000	4186	0.6	214
Ethanol	785.1	2440	0.17	109
Ice	920	2090	~ 2	150
Glass	2500	840	1.05	27
Aluminum	2700	900	238	72
Copper	8600	387	397	51
Iron	7800	448	79.5	33
Lead	11300	128	34.7	87

Latent Heat of Materials

	$\Sigma Q_{\text{iso}} = 0$	$\Delta Q = mc\Delta T$	$\Delta Q = L\Delta m$
Material	Melting ($^\circ\text{C}$)	L_{fusion} (J/kg)	Boiling ($^\circ\text{C}$) L_{vapor} (J/kg)
Nitrogen	-209.97	2.55×10^4	-195.81 2.01×10^5
Water	0.00	3.33×10^5	100.0 2.26×10^6
Ethanol	-114	1.04×10^5	78 8.54×10^5

Physics - Thermodynamics